

Electromagnetic Theory

(EE3005)

Date: September 24, 2024

Course Instructor(s)

1. Mohsin Yousuf (Course Moderator)

2. Dr. Huzaifa Rauf

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Roll No

Section

Student Signature

Do not write below this line

1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 01: Demonstrate the use of 3D orthogonal coordinate system and vector analysis tools (curl, divergence, etc.) in problem solving

Q1: The Fig.1 shown ahead is a 3D image drawn using values $0 \leq r \leq 1, 0 \leq \theta \leq \frac{\pi}{4}, 0 \leq \phi \leq \frac{7\pi}{4}$ (solid) [20 marks]

image drawn using values (mentioned at the top of the figure) from spherical coordinate system (SCS). Using differential method, find:

- (a) the total volume of the image,
- (b) the total surface area of the image,
- (c) the length of the circular edge as mentioned in the Fig. 1.

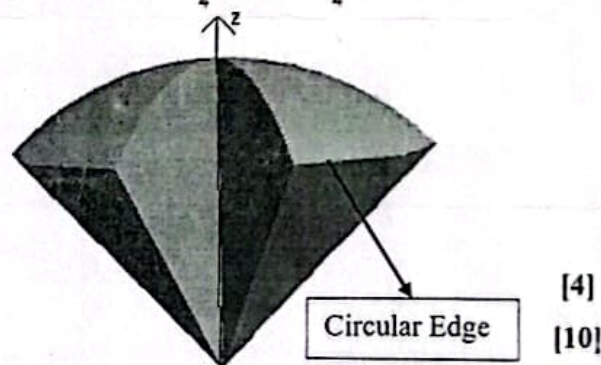


Figure 1. A 3D image with circular edge

- (d) Also, convert the point $P \left(r = 1\text{m}, \theta = \frac{\pi}{4}, \phi = \frac{7\pi}{4} \right)$ into RCS. [3]

National University of Computer and Emerging Sciences

Lahore Campus

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q2: A uniform line charge is distributed along the z -axis from $z = 5\text{m}$ to ∞ and from $z = -5\text{m}$ to $-\infty$ as shown in the Fig. 2 with $\rho_L = 20 \text{ [nC/m]}$. There is no line charge from $z = -5$ to 5m .

[20 marks]

Formulate the problem for dE in RCS and compute the total electric field intensity, E , at $(2, 0, 0)\text{m}$ in CCS.

Hint: Finding symmetry in the problem may reduce the components you need to find.

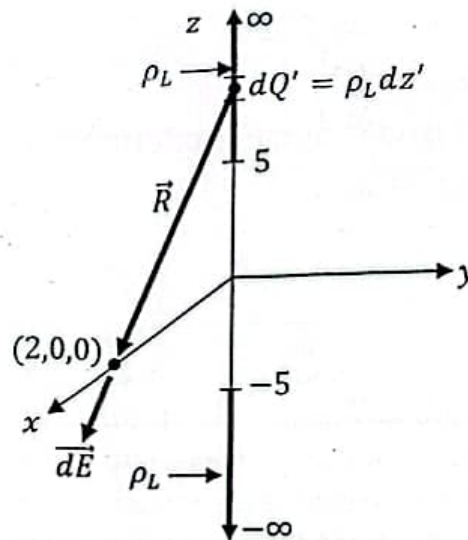


Figure 2. Two semi-infinite line charges on z -axis

Hint:

$$\int \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

and

$$\int \frac{x dx}{(a^2 + x^2)^{3/2}} = \frac{-1}{\sqrt{a^2 + x^2}}$$

Electromagnetic Theory (EE3005)

Final Exam

Date: January 01, 2025

Course Instructor(s)

1. Mohsin Yousuf (Course Moderator)
2. Dr. Huzaifa Rauf

Total Time (Hrs): 3
Total Marks: 100
Total Questions: 5

22L-6234

Roll No

5A

Section

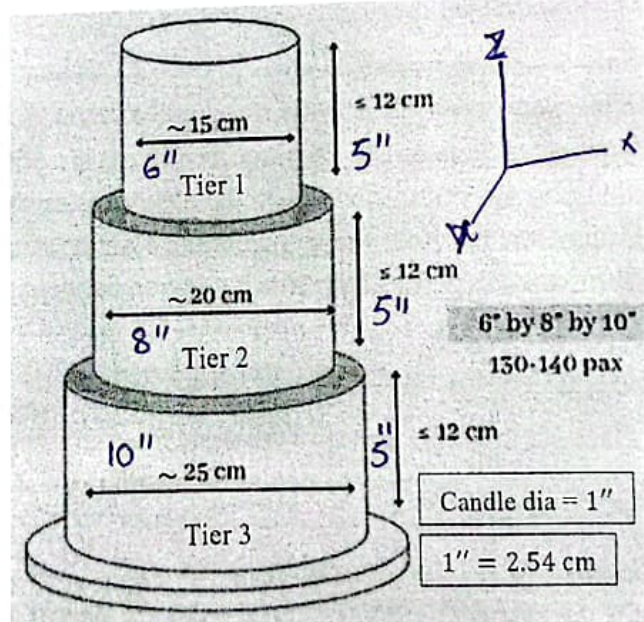
Haseeb
Student Signature

Do not write below this line

1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 01: Demonstrate the use of 3D orthogonal coordinate system and vector analysis tools in problem solving

Q1: Sister Ayesha is good in baking cakes but she is often confused about the dimensions of wrapping and the count of candles to lit on the cake. Now, she is up to making a big 3-tier cake for an upcoming event. The diameter of each tier in inches are mentioned as 6" $\times$ 8" $\times$ 10". The height of each tier is 12 cm. Help her in decorating by answering the following.



[20 marks]

- (a) Identify the coordinate system and determine the total volume (in cm^3) of the 3-tier cake using differential method only. [8]
- (b) Determine the surface area (in cm^2) and the length (in cm) of the ribbon required to wrap each tier up to height of 10 cm using differential method only. [6]
- (c) Determine the count of candles required to cover the sides (shaded as gray) of Tier 2 and Tier 3 top surfaces without gap. [6]

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

[20 marks]

Q2: A precision sensor system for a high-tech device, operating in a region with spatially varying dielectric properties is to be designed. The electric potential in the field is given by:

$$V = 20x^2yz - 10z^2 \quad (\text{in free space})$$

However, the region is partially filled with a dielectric material where the permittivity ϵ_r is function of the space and varies as:

$$\epsilon_r(y, z) = 1 + 0.5y^2z$$

- (a) Derive the equations for the conducting surfaces when $V = 0$ and $V = 60$ V. [3]
- (b) At point $P(2, y, 1)$, determine 'y' when the surface potential is $V = 60$ V. [2]
- (c) Formulate for electric field intensity E [V/m] and the electric flux density D [nC/m²] at $P(2, y, 1)$ considering the dielectric's spatial variation. Use value of 'y' as already found in part (b) in calculating E_P and D_P . [8]
- (d) Determine a unit vector 'n' normal to the $V = 60$ V equipotential surface at $P(2, y, 1)$ and directed towards the $V = 0$. [3]
- (e) Knowing the property of a perfect conductor, we can safely assume that the $V = 60$ V surface is a perfect conductor. Determine the surface charge density, ρ_s [nC/m²] at $P(2, y, 1)$, using two different methods. [4]

CLO # 03: Formulate the capacitance with one dimensional potential variation using direct integration.

Q3: A parallel-plate capacitor is made using two circular plates each of radius a , with the bottom plate on the xy plane, centred at the origin. The top plate is located at $z = 4\pi$, with its center on the z axis. Given the potential; $V_0 = 100 e^\pi$ is on the top plate while the bottom plate is grounded. Dielectric having radially dependent permittivity fills the region between plates is given by:

$$\epsilon(\rho) = \epsilon_0 \left(1 + \frac{\rho^2}{a^2} \right)$$

Formulate for the capacitance of the system by utilizing one-dimensional change in potential and direct integration method by answering the following:

- (i) Draw the diagram and solve for the potential; $V(z)$ using initial boundary conditions. [2+6]
- (ii) Solve for the electric field intensity; E . And the electric flux density; D . [4]
- (iii) Solve for the surface charge density, ρ_s and the total charge; Q [6]
- (iv) Compute the capacitance, C . [2]

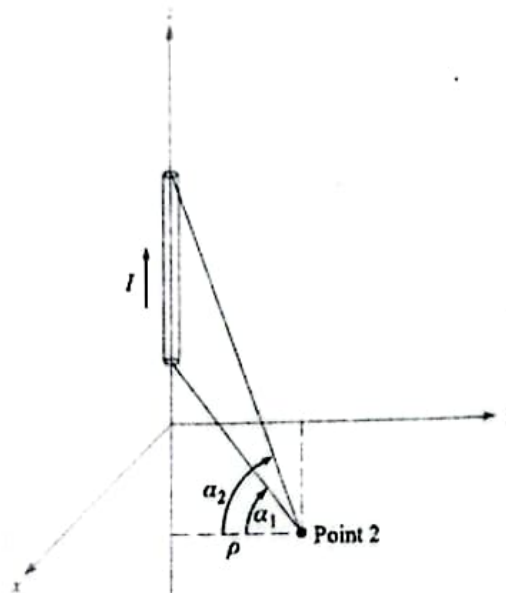
Note: Solve the problem using Laplace's Equation in Cylindrical Coordinates exclusively. No other method or approach will be considered for marking.

CLO # 04: Formulate magnetic fields and its effects for a given distribution of moving charges using laws of magnetostatics.

Q4:

[20 marks]

- (a) State Biot-Savart's law and mathematically express it using diagram. Also write different forms of the law for line, surface, and volume current. **[4]**
- (b) Formulate magnetic field intensity at Point 2 shown below. **[6]**



- (c) Suppose we have three current sheets, $K_1 = 2.7 \mathbf{a}_x$ [A/m] at $y = 0.1$, $K_2 = -1.4 \mathbf{a}_x$ [A/m] at $y = 0.15$, and $K_3 = -1.3 \mathbf{a}_x$ [A/m] at $y = 0.25$. Determine the value of $\mathbf{H}$ in RCS everywhere by showing complete working with the help of a suitable diagram. **[10]**

Regions $\rightarrow$	$-\infty < y \leq 0.1$	$0.1 \leq y \leq 0.15$	$0.15 \leq y \leq 0.25$	$y > 0.25$
-----------------------	------------------------	------------------------	-------------------------	------------

CLO # 04: Formulate magnetic fields and its effects for a given distribution of moving charges using laws of magnetostatics.

Q5:

[20 marks]

- (a) A cylindrical conductor of radius $a = 0.05$ m carries a uniformly distributed current $I = 10$ A in the +ve $\mathbf{a}_z$ direction. The magnetic field intensity $\mathbf{H}$ inside the conductor ($0 \leq \rho \leq a$) is given as $H_\phi = \frac{I\rho}{2\pi a^2}$. **[12]**
- (i) Verify Ampere's circuital law by calculating $\oint \mathbf{H} \cdot d\mathbf{L}$ for a circular path of radius $\rho = 0.03$ m.
- (ii) Compute the curl, $\nabla \times \mathbf{H}$ at $\rho = 0.03$ m and verify that it matches the current density $\mathbf{J}$ there.
- (b) A circular loop located on $x^2 + y^2 = 16$, $z = 0$ carries a direct current of 5 A along $\mathbf{a}_\phi$. Formulate $\mathbf{H}$ at $(0, 0, 5)$ and $(0, 0, -5)$. **[8]**

National University of Computer and Emerging Sciences Lahore Campus

Formula Sheet

$$F_1 = -F_2 = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{12} = -\frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \mathbf{a}_{12} \quad E = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad E(\mathbf{r}) = \sum_{m=1}^n \frac{Q_m}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_m|^2} \mathbf{a}_m \quad E = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_N$$

$$E = \frac{\rho_L}{2\pi\epsilon_0 R} \mathbf{a}_\rho \quad D = \int_{\text{vol}} \frac{\rho_v dV}{4\pi R^2} \mathbf{a}_R \quad \Psi = \oint_S \mathbf{D}_s \cdot d\mathbf{S} = \text{charge enclosed} = Q \quad \mathbf{J} = \rho_v \mathbf{v} \quad \text{div } \mathbf{D} = \rho_v$$

$$\text{div } \mathbf{D} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \quad \text{div } \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z} \quad (\text{cylindrical}) \quad E = -\nabla V$$

$$\text{div } \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta D_\theta) + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi} \quad (\text{spherical}) \quad W = -Q \int_{\text{init}}^{\text{final}} E \cdot d\mathbf{L} \quad \nabla^2 V = 0$$

$$d\mathbf{L} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z \quad (\text{rectangular})$$

$$d\mathbf{L} = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z \quad (\text{cylindrical})$$

$$d\mathbf{L} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi \quad (\text{spherical})$$

$$V(\mathbf{r}) = \int_{\text{vol}} \frac{\rho_v(\mathbf{r}') dV'}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|}$$

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z$$

$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi$$

$$I = \int_S \mathbf{J} \cdot d\mathbf{S} \quad (\nabla \cdot \mathbf{J}) = -\frac{\partial \rho_v}{\partial t}$$

$$W_E = \frac{1}{2} \int_{\text{vol}} \mathbf{D} \cdot \mathbf{E} dV = \frac{1}{2} \int_{\text{vol}} \epsilon_0 E^2 dV$$

$$W_E = \frac{1}{2} \int_{\text{vol}} \rho_v V dV \quad \mathbf{J} = \sigma \mathbf{E}$$

$$\mathbf{E} \times \mathbf{n}|_s = 0$$

$$\sigma = -\rho_s \mu_e$$

$$\mathbf{D} \cdot \mathbf{n}|_s = \rho_s$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2} \quad P = \lim_{\Delta v \rightarrow 0} \frac{1}{\Delta v} \sum_{i=1}^N P_i$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \mathbf{P} = \chi_e \epsilon_0 \mathbf{E}$$

$$(\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{n} = \rho_s \quad (\mathbf{E}_1 - \mathbf{E}_2) \times \mathbf{n} = 0$$

$$\mathbf{H} = \oint \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} \quad W_E = \frac{1}{2} C V_0^2 = \frac{1}{2} Q V_0 = \frac{1}{2} \frac{Q^2}{C}$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon}$$

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \left(\frac{\partial^2 V}{\partial \phi^2} \right) + \frac{\partial^2 V}{\partial z^2} \quad \oint \mathbf{H} \cdot d\mathbf{L} = I \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{S}$$

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

$$I d\mathbf{L} = \mathbf{K} d\mathbf{S} = \mathbf{J} dV$$

$$\nabla \times \mathbf{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \mathbf{a}_\phi + \left(\frac{1}{\rho} \frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{1}{\rho} \frac{\partial H_\rho}{\partial \phi} \right) \mathbf{a}_z \quad (\text{cylindrical})$$

	$\mathbf{a}_r$	$\mathbf{a}_\theta$	$\mathbf{a}_\phi$
$\mathbf{a}_x$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\mathbf{a}_y$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$\cos \phi$
$\mathbf{a}_z$	$\cos \theta$	$-\sin \theta$	0

$$V = \frac{Q}{4\pi\epsilon_0 r} \quad \mathbf{B} = \mu_0 \mathbf{H}$$

$$\nabla \times \mathbf{H} = \frac{1}{r \sin \theta} \left(\frac{\partial (H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \mathbf{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial (r H_\phi)}{\partial r} \right) \mathbf{a}_\theta + \frac{1}{r} \left(\frac{\partial (r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \mathbf{a}_\phi \quad (\text{spherical})$$

$$\mathbf{H} = \frac{I}{4\pi\rho} (\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_\phi$$

$$\mathbf{F} = \oint I d\mathbf{L} \times \mathbf{B}$$

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

$$\frac{dW_E}{dV} = \frac{1}{2} \mathbf{D} \cdot \mathbf{E}$$

$$\int \frac{1}{(a^2 \pm x^2)^{3/2}} dx = \frac{\pm x}{a^2 \sqrt{a^2 \pm x^2}} \int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$dS_\rho = \rho d\phi dz \mathbf{a}_\rho, dS_\phi = \rho dz d\phi \mathbf{a}_\phi, dS_z = \rho d\phi d\rho \mathbf{a}_z, dV = \rho d\rho d\phi dz$$

$$dS_r = r^2 \sin \theta d\theta d\phi \mathbf{a}_r, dS_\theta = r \sin \theta dr d\phi \mathbf{a}_\theta, dS_\phi = r dr d\theta \mathbf{a}_\phi$$

$$dV = r^2 \sin \theta dr d\theta d\phi$$

Electromagnetic Theory (EE3005)

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Date: November 5, 2024

Course Instructor(s)

1. Mohsin Yousuf (Course Moderator)
2. Dr. Huzaifa Rauf

22L-6234
Roll No

5A
Section

Haseeb
Student Signature

Do not write below this line

1. Attempt all questions and remember to solve parts of the same question together.
2. Final answers should be correct up to two decimal places with proper SI units.
3. Show all the steps with the help of diagrams and equations.

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q1: In a region of free space, the electric flux density D is defined by: [20 marks]

$$D = \begin{cases} \rho_0(z + 2d) \mathbf{a}_z & \text{C/m}^2 \text{ for } -2d \leq z \leq 0, \\ -\rho_0(z - 2d) \mathbf{a}_z & \text{C/m}^2 \text{ for } 0 \leq z \leq 2d, \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) Define Maxwell's first equation (both in point and integral form) and find the volume charge density ρ_v everywhere. [5]
- (b) Determine the electric flux ψ passing through the surface along $+\mathbf{a}_z$, defined by $z = 0, -a \leq x \leq a, -b \leq y \leq b$. [5]
- (c) Formulate for the total charge enclosed within the following regions: [10]

i) Region 1:

$$-a \leq x \leq a,$$

$$-b \leq y \leq b,$$

$$-d \leq z \leq d$$

call it $Q_{\text{encl},1}$

ii) Region 2:

$$-a \leq x \leq a,$$

$$-b \leq y \leq b,$$

$$0 \leq z \leq 2d$$

call it $Q_{encl,2}$.

CLO # 02: Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.

Q2: Answer the following:

[20 marks]

(a) Define electric dipole and dipole moment $\mathbf{p}$ with the help of diagram. [4]

(b) A dipole of moment $\mathbf{p} = 3\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z$ [nC · m] is located at the origin in free space. Find V at $P(r = 3.5 \text{ m}, \theta = 30^\circ, \phi = 60^\circ)$. [6]

Hint,
$$V = \frac{\mathbf{p} \cdot \mathbf{a}_r}{4\pi\epsilon_0 r^2}$$

(c) Four point-charges $Q_1 = +2q$, $Q_2 = -q$, $Q_3 = -q$, and $Q_4 = +2q$ are located at $(0, 0)$, $(a, 0)$, $(0, a)$, and (a, a) respectively in yz -plane. Formulate for the potential energy stored in this point charge constellation. [10]

Hint,
$$W_E = \frac{1}{2} \sum Q_i V_i$$

